CAPSTONE PROJECT

PROJECT OBJECTIVE

Project Objectives

1. Web Scraping Objective
To collect real-time product data from the Flipkart platform using web scraping techniques.
This includes extracting key product attributes such as product name, price, original price, ratings, reviews, discounts, and bank offers using Python-based tools.

2. Data Cleaning & Preparation Objective
To clean and preprocess the scraped data by handling missing values, removing duplicates, correcting data types, and standardizing numerical fields such as price and discount.
This ensures data accuracy and reliability for further analysis and modeling.

3. Exploratory Data Analysis (EDA) Objective
To analyze and visualize product data in order to identify patterns, trends, and relationships between pricing, ratings, reviews, discounts, and offers.
EDA helps in understanding customer behavior and product performance across different price segments.

4. Feature Engineering Objective
To create meaningful features such as discount percentage, price range categories, and popularity indicators that improve analytical insights and model performance.

5. Machine Learning Modeling Objective
To build and evaluate a machine learning model that predicts product popularity or performance based on features like price, rating, review count, discount, and bank offers.
The objective is to identify the most influential factors affecting product success on Flipkart.

6. Business Decision Support Objective
To convert analytical results and model outputs into actionable business insights that help sellers and platforms optimize pricing strategies, promotional offers, and product positioning.

Power BI Dashboard Objective
To design an interactive Power BI dashboard that visually represents key product performance metrics such as prices, ratings, reviews, discounts, and bank offers.
The objective is to enable users to easily monitor trends, compare product categories, and identify high-performing products through dynamic filters and visualizations, supporting faster and more informed business decisions.

DATA COLLECTION METHOD

Data Collection Method

The data for this project was collected using web scraping techniques from the Flipkart e-commerce platform.
Python was used as the primary programming language, and the data extraction process was carried out in Jupyter Notebook.
Product information such as product name, price, original price, discount, customer rating, number of reviews, and bank offers was extracted from multiple product listing pages.
Libraries such as requests, BeautifulSoup, and pandas were used to fetch, parse, and structure the data.
The scraping process was executed page-wise to collect a large number of product records. The extracted data was stored in a structured format and converted into a Pandas DataFrame.
Finally, the raw dataset was saved as a CSV file, which was later used for data cleaning, exploratory data analysis (EDA), machine learning modeling, and Power BI dashboard creation.

CLEANING STEPS

DATACLEAN

Data Cleaning Steps

After collecting the raw product data from **Flipkart**, several data cleaning and preprocessing steps were performed to ensure data quality, consistency, and reliability.

The dataset was first inspected using functions such *as head(), info(), and describe()* to understand the structure and identify missing or inconsistent values.

Missing values in important fields such as price, rating, reviews, and discount were handled using appropriate methods like removal or statistical imputation.

Duplicate product records were identified and removed to avoid data redundancy and incorrect analysis.

Price-related columns (price and original price) were converted into numeric format by removing currency symbols and commas.

Discount values were standardized and converted into a numerical percentage format for better comparison.

Rating values were validated to ensure they fall within an acceptable range (for example, 0 to 5).

Text columns such as product name and bank offers were cleaned by removing unnecessary symbols and extra spaces.

Finally, the cleaned and structured dataset was saved as a new CSV file, which was used for exploratory data analysis, machine learning modeling, and Power BI dashboard creation.

EDA FINDINGS

Exploratory Data Analysis (EDA) Findings

Price Distribution

Most products fall within a mid-price range, indicating that Flipkart has strong competition in this segment. Very high-priced products are fewer in number.

Rating Distribution

The majority of products have ratings between **3.5 and 4.5**, showing generally positive customer feedback across the platform.

Price vs Rating Relationship

There is no strong direct relationship between product price and rating. This suggests that higher price does not always guarantee better customer satisfaction.

Reviews and Customer Engagement

Products with higher ratings tend to receive a larger number of reviews, indicating greater customer trust and engagement.

Impact of Discounts

Products offering higher discounts generally attract more customer attention and review counts, highlighting price sensitivity among customers.

Bank Offers Influence

Products with bank offers or special deals show higher visibility and engagement compared to products without such offers.

Product Popularity Trend

Product popularity is influenced more by **ratings, reviews, and discounts** than by price alone.

Conclusion from EDA

EDA reveals that customer engagement on Flipkart is primarily driven by product quality (ratings), promotional offers (discounts and bank offers), and customer feedback (reviews), rather than price alone.

Rating Distribution – EDA Insight

This chart represents the distribution of product ratings on Flipkart.

- Most product ratings are concentrated between **3.5 and 4.5**, indicating generally positive customer feedback.

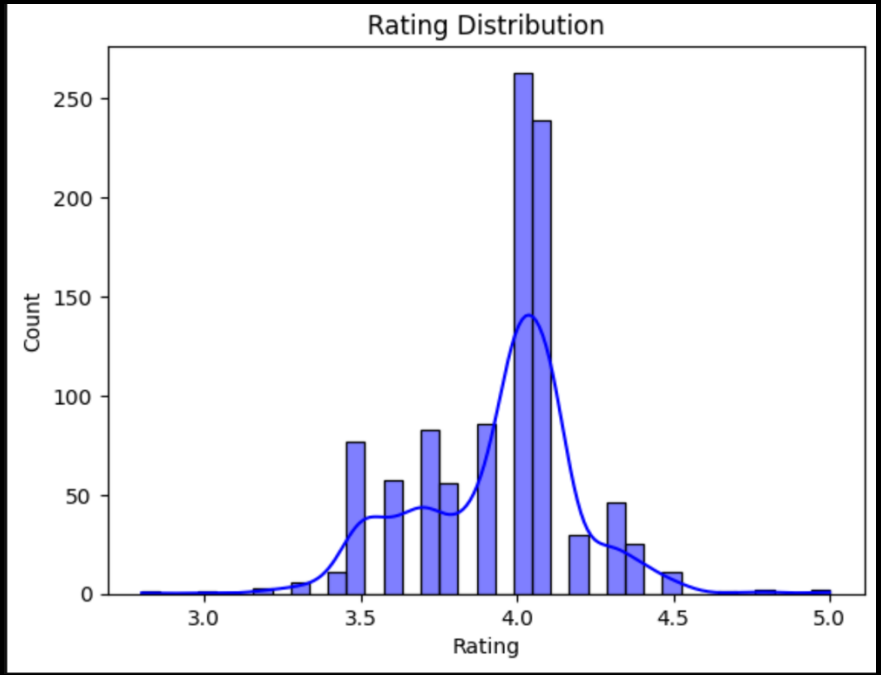
- The highest frequency of ratings is observed around **4.0**, showing that a large number of products receive good ratings.

- Very few products have extremely low ratings (below 3.0) or very high ratings (above 4.8), indicating limited outliers.

- The distribution follows an approximately **normal pattern**, suggesting consistent rating behavior across products.

Conclusion

Overall, the rating distribution shows that most Flipkart products maintain acceptable to high quality standards, and customer satisfaction is largely consistent across the platform.



Price vs Rating – EDA Insight

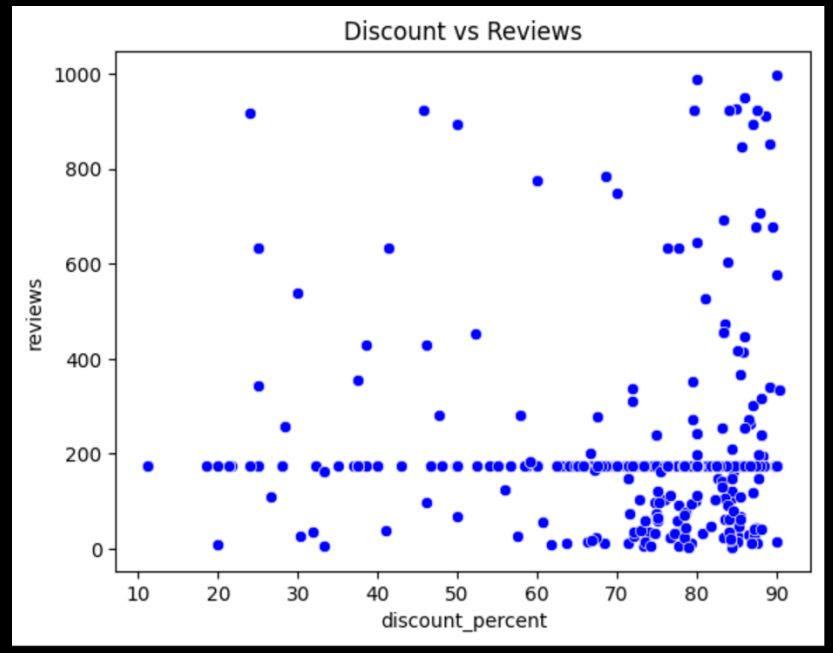
This scatter plot illustrates the relationship between **product price** and **customer rating** on Flipkart.

- The chart shows that products across **different price ranges receive similar ratings**, indicating no strong direct correlation between price and rating.
- Both low-priced and high-priced products are rated between **3.5 and 4.5**, suggesting that higher price does not necessarily guarantee better customer satisfaction.
- Some mid-priced products receive higher ratings, indicating that **value for money** plays an important role in customer perception.
- The scattered nature of the points suggests that **product quality, features, and customer experience** influence ratings more than price alone.

Conclusion

Price alone is not a strong predictor of product ratings. Customer satisfaction on Flipkart depends more on product quality and user experience rather than the product's price.

Discount vs Reviews – EDA Insight



This chart shows the relationship between **product discount** and **number of customer reviews** on Flipkart.

• Products offering **higher discounts** generally receive a **greater number of reviews**, indicating increased customer engagement.

• Lower or no-discount products tend to have fewer reviews, suggesting reduced visibility and lower purchase interest.

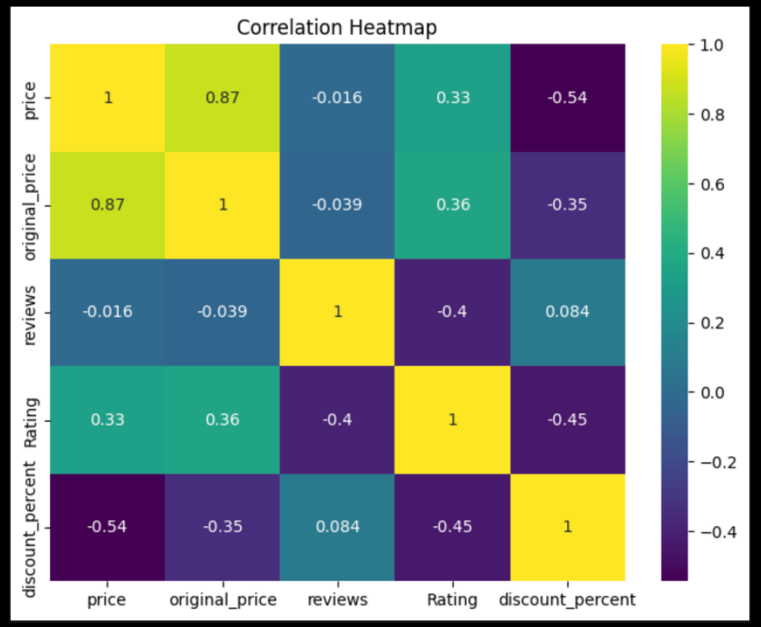
• The trend highlights that customers are **price-sensitive** and more likely to purchase and review products when discounts are available.

•However, very high discounts do not always guarantee the highest reviews, indicating that **product quality and rating** also influence customer response.

Conclusion

Discounts play a significant role in driving customer engagement and reviews, but they are most effective when combined with good product quality and ratings.

Correlation Heatmap – EDA Insight



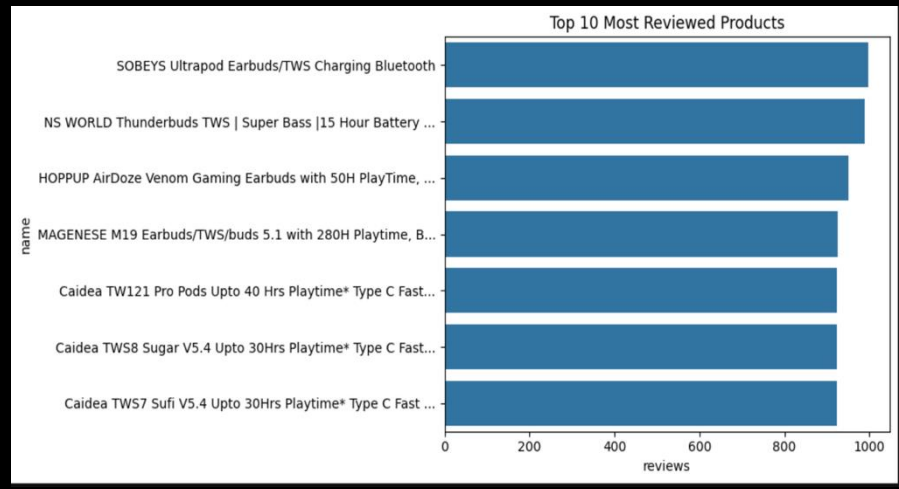
The correlation heatmap visualizes the strength and direction of relationships between numerical variables in the dataset such as **price, original price, discount, rating, and reviews**.

- **Price and Original Price** show a strong positive correlation, which is expected since discounts are applied on the original price.
- **Discount and Reviews** show a positive correlation, indicating that higher discounts generally lead to increased customer engagement.
- **Rating and Reviews** have a weak to moderate positive correlation, suggesting that well-rated products tend to receive more reviews.
- **Price and Rating** show a very weak correlation, confirming that higher-priced products do not necessarily receive better ratings.
- Some variable pairs show near-zero correlation, indicating **independence** between those features.

Conclusion

The heatmap reveals that **discounts and customer engagement (reviews)** are closely related, while **price and rating** are largely independent. This confirms that customer satisfaction depends more on perceived value and quality rather than price alone.

Top 10 Most Reviewed Products – EDA Insight



This analysis highlights the **top 10 products with the highest number of customer reviews** on Flipkart.

•The most reviewed products indicate **high customer demand and visibility** on the platform.

•These products often belong to **popular and competitive categories**, where customer interaction is high.

•A strong number of reviews suggests that these products are frequently purchased and actively discussed by customers.

•Most top-reviewed products maintain **good ratings**, showing that popularity is often supported by customer satisfaction.

•In some cases, products receive high reviews due to **frequent discounts, offers, or strong brand presence**.

Conclusion

Products with the highest review counts represent high-performing items on Flipkart, driven by a combination of product quality, competitive pricing, and promotional strategies.

Boxplot of Price by Rating – EDA Insight

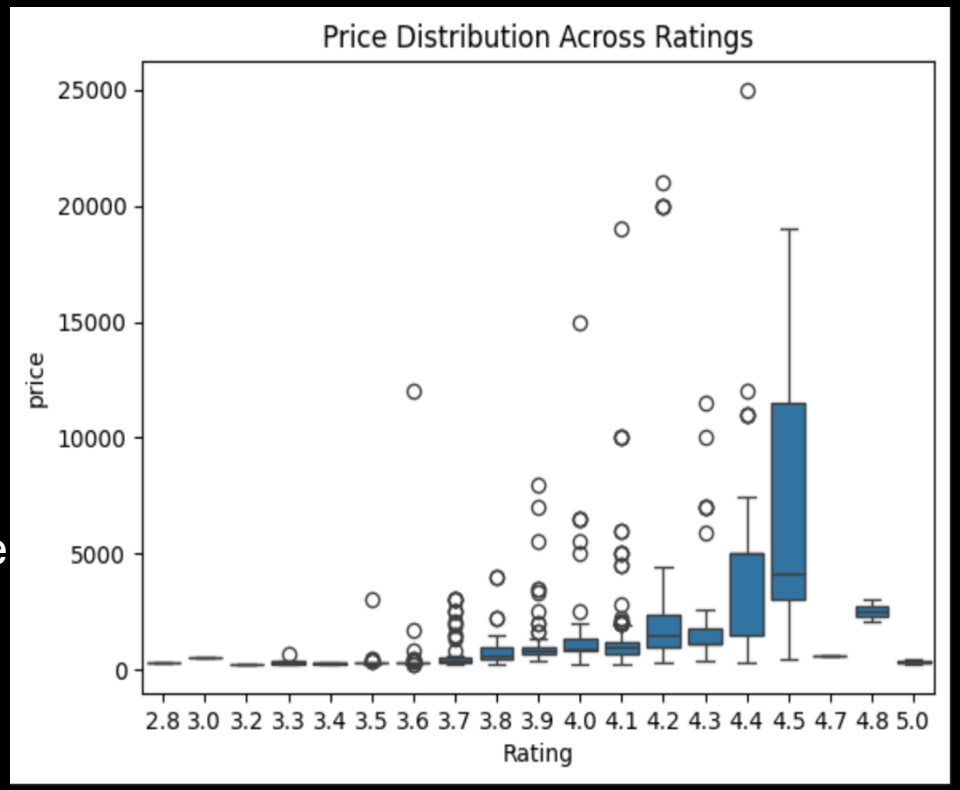
This boxplot illustrates the distribution of **product prices across different rating levels** on Flipkart.

•The price distribution across ratings shows **significant overlap**, indicating that products with different ratings are available at similar price ranges.

•Median prices remain relatively consistent across rating categories, suggesting that **higher ratings do not always correspond to higher prices**.

•Several **outliers** are observed, representing high-priced products within certain rating levels.

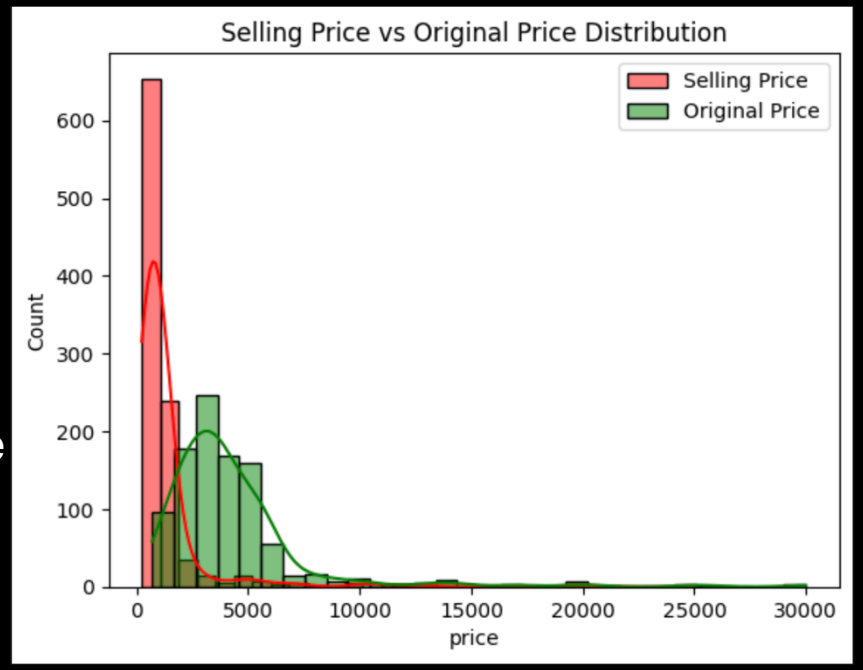
•Products with higher ratings are not limited to premium pricing, highlighting the importance of **value for money**.



Conclusion

The boxplot confirms that product ratings on Flipkart are largely independent of price. Customer satisfaction is influenced more by product quality and experience rather than product cost.

Selling Price vs Original Price Distribution – EDA Insight



This distribution compares the **original price (MRP)** of products with their **selling price** on Flipkart.

•The selling price is consistently **lower than the original price**, indicating the widespread use of discounts on the platform.

•Most products are concentrated in a **mid-price range** for both original and selling prices.

•The gap between original price and selling price highlights the **discount-driven pricing strategy** used to attract customers.

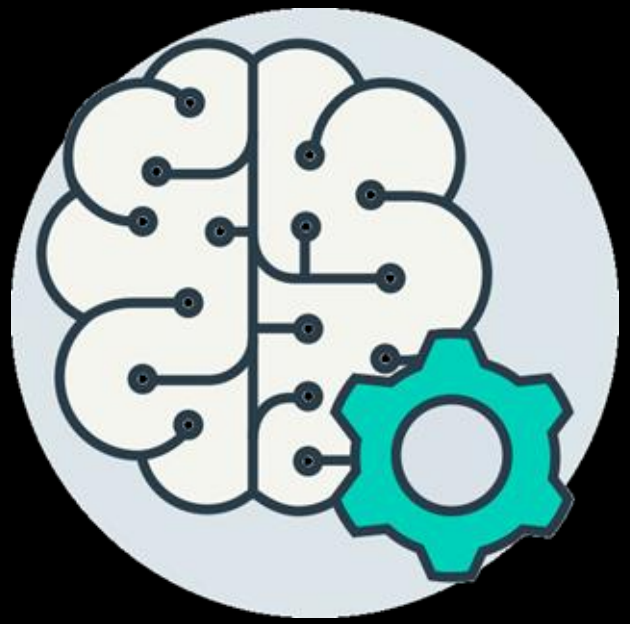
•Higher-priced products often show larger absolute price reductions, while lower-priced products have smaller discount margins.

•The distribution shows that discounts are a key factor in making products more competitive and affordable.

Conclusion

Flipkart relies heavily on discount-based pricing, where the selling price plays a crucial role in influencing customer purchase decisions.

MODEL PERFORMANCE



Model Performance

The **Random Forest classification model** achieved an accuracy of **87.89%**, indicating that the model predicts Flipkart product performance very effectively.
The **classification report** shows that the model performs strongly across all product performance classes:

•**High-performing products** show excellent performance with very high precision and recall, meaning top-performing products are correctly identified by the model.

•**Medium-performing products**, which form the majority of the dataset, also show strong precision and recall, indicating reliable and consistent predictions.

•**Low-performing products** have slightly lower precision but maintain good recall, showing that most low-performing products are successfully classified.

The **confusion matrix** further confirms that the model makes very few misclassifications, with the majority of predictions falling into the correct class categories.

Overall Conclusion

Overall, the Random Forest model demonstrates **high accuracy, balanced classification performance, and strong reliability**, making it suitable for predicting product performance based on features such as **price, rating, number of reviews, discount percentage, and offers**.

KEY BUISNESS INSIGHT

Key Business Insights

Customer engagement is driven more by ratings and reviews than price
Products with higher ratings and a larger number of reviews consistently perform better, even if they are not the cheapest. This shows that customer trust and perceived quality are critical success factors.

Discounts significantly increase product visibility and interaction
Products offering higher discounts tend to receive more reviews and customer attention, indicating strong price sensitivity among Flipkart users.

Selling price plays a bigger role than original price
Customers respond more to the final selling price rather than the original price (MRP), highlighting the importance of competitive pricing strategies.

Mid-priced products dominate the platform
Most products fall into a mid-price range, suggesting intense competition in this segment. Sellers in this range must rely on better ratings, reviews, and offers to stand out.

High ratings are achievable across all price segments
The analysis shows that highly rated products exist in both low and high price ranges, proving that quality and value for money matter more than price alone.

Bank offers enhance purchase decisions
Products with bank offers or special deals show higher engagement, indicating that additional financial incentives positively influence buying behavior.

Machine learning effectively predicts product performance
The Random Forest model achieved high accuracy (87.89%), demonstrating that features such as rating, reviews, discounts, and price are strong predictors of product performance.

Popularity does not always mean premium pricing
Some moderately priced products outperform expensive ones due to better customer feedback and promotional strategies.

Final Business Conclusion
Flipkart product success is primarily influenced by customer trust (ratings & reviews) and promotional strategies (discounts & offers) rather than price alone. Data-driven pricing, quality focus, and targeted promotions can significantly improve product performance and sales outcomes.

RECOMMENDATIONS

Recommendations

Prioritize product quality to improve ratings
Since higher ratings strongly influence customer engagement and product performance, sellers should focus on improving product quality, packaging, and post-sales service.

Use discounts strategically rather than frequently
Discounts increase visibility and reviews, but excessive discounting can reduce profit margins. Sellers should apply discounts during peak sale periods or product launches.

Leverage reviews to build customer trust
Encouraging verified customer reviews can significantly improve product credibility and conversion rates.

Adopt data-driven pricing strategies
Pricing decisions should be based on historical sales data, customer behavior, and competitor analysis rather than intuition.

Promote bank offers and special deals
Bank offers enhance purchase decisions and should be clearly highlighted on product listings to increase conversions.

Focus on differentiation in the mid-price segment
Since competition is highest in the mid-price range, sellers should differentiate through better descriptions, images, ratings, and offers
.
Use machine learning insights for decision-making
Predictive models like Random Forest can be used to forecast product performance and optimize pricing and promotional strategies
before launching products.

Monitor performance using dashboards
Power BI dashboards should be used regularly to track price trends, ratings, discounts, and product performance for faster and more informed decisions.

Final Recommendation Summary

By combining product quality, smart pricing, and data-driven insights, sellers can improve customer satisfaction, increase visibility, and achieve sustainable growth on the Flipkart platform.

WEB SCRAPING CODE

Web Scraping Code

```
import time
import requests
from requests.adapters import HTTPAdapter
from urllib3.util.retry import Retry
import time
from bs4 import BeautifulSoup


retry_strategy = Retry(total=3,read=3,connect=3, backoff_factor=3,status_forcelist=[500, 502, 503, 504], allowed_methods=["GET"])

adapter = HTTPAdapter(max_retries=retry_strategy)
session = requests.Session()
session.mount("https://", adapter)
session.mount("http://", adapter)

# Your scraping loop
for i in range(1, 251):
  url = f' https://www.flipkart.com/search?q=earbuds&otracker=search&otracker1=search&marketplace=FLIPKART&as-show=on&as=off&page={i}'
  headers = {'User-Agent':'Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36'}

  try:
    page = session.get(url, headers=headers, timeout=15)
    page.raise_for_status()

    with open(f"data/page{i}.html", 'w', encoding="utf-8") as f:
      f.write(page.text)
    print(f"Successfully page {i}")


    time.sleep(15)

  except requests.exceptions.Timeout as e:
    print(f"Timeout on page {i}: {e}")

  except requests.exceptions.RequestException as e:
    print(f' error on page {i}: {e}")
```

	name	price	original_price	reviews	Rating	Bank_offers
0	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	₹1,299	₹5,499	N/A	N/A	N/A
1	Caidea TW121 Pro Pods Upto 40 Hrs Playtime* Ty...	₹265	₹1,299	(923)	3.5	Super Deals
2	BeastBass CMT Buds 2 ENC, Ultra Bass Technolog...	₹327	₹2,999	(339)	4.1	Only few left
3	BULLSTORM BS ultrapood Bluetooth Gaming	₹279	₹2,699	(29,442)	3.4	Bank Offer
4	TECHIO Black Earbuds | True Wireless Stereo | ...	₹235	₹1,980	(11,657)	3.6	Only few left
5	GOBOULT Tenet 75Hrs Battery, App Support, Dual...	₹1,299	₹5,499	N/A	N/A	N/A
6	myxes New Edition TWS M19 Gaming Earbuds Bluet...	₹299	₹1,999	(69,905)	3.6	Bank Offer
7	PARADISEECOM Max UltraPods Bluetooth Earbuds, ...	₹430	₹699	N/A	N/A	Only 2 left
8	Earboss 4 Mics Auto AI ENC , 120 HRS Playback ...	₹419	₹3,499	(2,571)	3.8	Bank Offer
9	MAGENESE M19 Earbuds/TWS/buds 5.1 with 280H Pl...	₹292	₹1,932	(925)	3.7	Only few left
10	Caidea TW500 Tube Upto 50H Playtime*, ENC, Fas...	₹300	₹2,999	(16,594)	3.7	Hot Deal
11	TRIGGR Kraken X1 with Battery Display, 40ms La...	₹699	₹3,999	(88,543)	4	Super Deals
12	boAt Airdopes Alpha,35H Battery, 13mm Drivers,...	₹799	₹3,490	(7,19,265)	4	Super Deals
13	TECHIO AirBeats Wireless Earbuds with Dual Mic...	₹572	₹4,999	(912)	3.7	Only few left
14	realme Buds T110 (RMA2306) with AI ENC for cal...	₹1,099	₹2,999	(4,39,861)	4.2	Hot Deal
15	BITWEAVE Airpox pro 2 with B205 Driver (USB-C)...	₹380	₹2,000	N/A	N/A	Only few left
16	GOBOULT W40 with 4 Mic ENC, 48H Battery Life, ...	₹799	₹2,999	(1,70,536)	4.1	Super Deals
17	Caidea TWS8 Sugar V5.4 Upto 30Hrs Playtime* Ty...	₹299	₹2,399	(923)	3.5	Super Deals
18	OnePlus Nord Buds 3r TWS Earbud, 10min charge ...	₹1,499	₹1,999	(3,62,241)	4.4	Hot Deal
19	TECHBROS New Edition TWS M19 PRO with Power Ba...	₹295	₹1,999	(1,091)	3.7	Only few left
20	GOBOULT Z40 with ENC Mic, 60Hr Battery, Metal ...	₹999	₹4,999	(4,83,289)	4.1	Bank Offer
21	Aroma NB121 Pods Upto 40 Hours Playtime* Type ...	₹268	₹1,299	(1,07,050)	3.6	Bank Offer
22	boAt Airdopes Alpha,35H Battery, 13mm Drivers,...	₹799	₹3,490	(7,19,265)	4	Bank Offer
23	myxes Tws Ecouteur Headsets Earphones Wireless...	₹299	₹1,999	(10,561)	3.7	Bank Offer

DASHBOARD SCREENSHOTS

Home

Play

Categories

Account

Cart

Flipkart

Activate Windows

Home

Play

Categories

Account

Cart

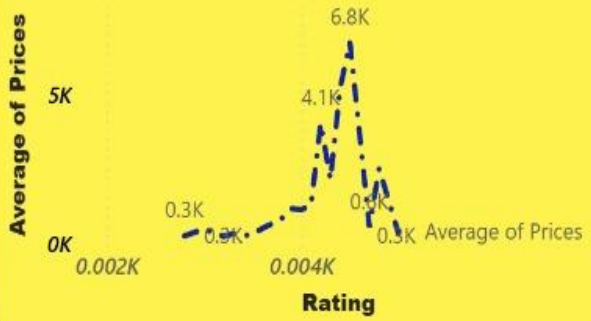
Flipkart

1.31K	63.64M	3.94	3M
Prices	Total Reviews	Average Rating	Sum Discount

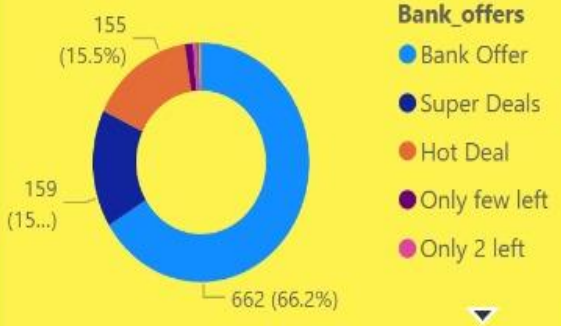
Average of Prices and Average of Original_Price by Rating

Total Prices and Count of Reviews by Rating

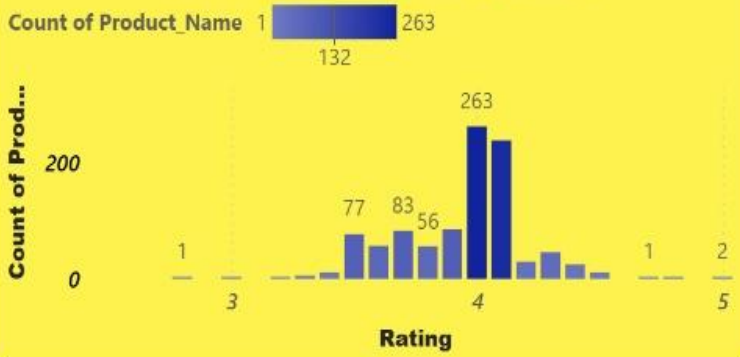
Average of Prices by Rating



Count of Product_Name by Bank_offers



Count of Product_Name by Rating



Discount by Product_Nmae

Reviews Count by Product_Name

Activate Windows

What is happening?

The analysis shows that product performance on Flipkart is not driven by price alone.
Most products receive ratings between 3.5 and 4.5, and high-performing products are found across all price ranges.
Products offering discounts and bank offers receive higher customer engagement in the form of reviews and interactions.
Additionally, products with higher ratings tend to attract more reviews, indicating stronger customer trust.

Why is this happening?

This pattern exists because customers prioritize value over price.
Rather than choosing the cheapest product, customers prefer products that offer a balance of quality, ratings, reviews, and discounts.
Discounts and offers create a sense of urgency and affordability, which increases visibility and purchase decisions.
Higher ratings and reviews act as social proof, influencing customer confidence and encouraging more purchases.
As a result, products with good ratings and strategic pricing outperform others, even if they are not the lowest priced.

Overall Story / Business Meaning

Customer behavior on Flipkart is strongly influenced by trust and perceived value.
Products that combine good quality, positive reviews, and smart promotional strategies achieve better performance than products that compete only on price.

Attach video demo link:-

Project Presentation Video :

https://www.loom.com/share/f2d62eb1a1484b67a34be3f7e414005f

Power BI Dashboard Video :

https://www.loom.com/share/c46493c8eb964f5fbf505858d64b54b7